CAMPUS-BRAIN

Project documentation

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Abstract:

Campus-Brain is a web-based academic learning platform designed for students and faculty. It provides an organized way to access and manage educational resources. The platform focuses on learning materials related to CSE. It provides separate dashboards for students and faculty members. Users can register, log in, and select their respective roles.

Students can view available learning resources and study materials. A search feature helps students find notes and materials easily. Students can upload and download files.

Faculty members can upload notes, presentations, and PDF documents. Faculty can view and manage uploaded learning resources. They can also view student requests for study materials. The platform provides a simple and user-friendly interface for academic resource management.

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Introduction:

In the present educational environment, students and faculty regularly use different sources to access academic information. Finding the required notes, subject materials,

and study resources can sometimes be difficult when the information is scattered across different platforms.

CampusBrain is designed to address this problem by providing an organized platform for academic resource management. It focuses on making learning materials easier to access and helping students prepare for their subjects more conveniently.

The project provides a structured interface through which users can access features according to their roles. It aims to simplify the process of finding, organizing, and managing educational resources while providing a convenient user experience.

CampusBrain is developed as a web-based prototype to demonstrate its main functionalities and user navigation. It provides a foundation for developing a more advanced academic resource management platform in the future.

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Problem Statement:

Students often face difficulties finding and organizing academic study materials because resources are available across different platforms. Faculty members may also find it challenging to manage and share learning materials efficiently. For example in whatsapp group if any teachers send pdf of particular topic, after few months it is

difficult to search that particular pdf.

Therefore, CampusBrain aims to provide an organized platform where students can access study resources and faculty can upload and manage educational materials in one place.

OBJECTIVES OF THE PROJECT:

The main objectives of CampusBrain are:

1. To provide an organized platform for academic learning materials.

2. To help students easily search for and access study resources.

3. To enable faculty members to upload and manage educational materials.

4. To provide separate dashboards for students and faculty.

5. To implement basic registration, login, profile management, and logout features.

6. To develop a simple and user-friendly interface for academic resource management.

7. To demonstrate the basic workflow of an academic learning platform through a web-based prototype.

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TECHNOLOGY STACK:

CampusBrain is currently developed as a frontend prototype using HTML5, CSS3, and JavaScript. The prototype is hosted through GitHub Pages and demonstrates the proposed user interface and basic interactions.

The project is maintained in the GitHub repository using the `master` branch. Future development will extend the prototype into a full-stack web application using Java, Spring Boot, and MySQL.

Technologies Used

* HTML5: Used to structure the web pages.

* CSS3: Used to design an attractive and responsive user interface.

* JavaScript: Used to implement frontend interactions such as searching, filtering, and navigation.

* Visual Studio Code: Used as the development environment.

* Git and GitHub: Used for version control and source code management.

* GitHub Pages: Used to host the frontend prototype.

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Planned Technologies for Future Development

* Java: For implementing backend application logic.

* Spring Boot: For developing backend services and APIs.

* MySQL: For storing user accounts and academic resource information.

* Maven: For managing backend dependencies and building the application.

The future implementation will integrate the frontend, backend, and database to provide a complete campus knowledge management system.

Software Requirements:

- Operating System: Windows, Linux, or macOS

- Web Browser: Google Chrome or Microsoft Edge

- Front-End Technologies: HTML, CSS, and JavaScript

- Hosting Platform: GitHub Pages

SYSTEM FLOWCHART:

The system flowchart illustrates the workflow of the CampusBrain web application, from opening website to

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logging out.

The process begins when the user opens CampusBrain. The user can sign up by selecting a role and entering the required details, or log in using an existing account. After successful login, the user accesses the appropriate dashboard based on their role.

Students can view and search for learning resources, while faculty members can upload and manage study materials and view student requests. Both users can view and edit their profile details. The process ends when the user logs out of the application.

The flowchart provides a visual representation of the system's navigation and major functions.

MODULE DESCRIPTION:

CampusBrain consists of the following modules:

1. User Authentication Module

This module allows users to register, log in, and access the application according to their selected role.

2. Student Module

This module allows students to view learning resources, search for study materials, and upload or download files for personal use in the prototype.

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3. Faculty Module

This module enables faculty members to upload learning materials, manage uploaded files, and view student requests for study materials.

4. Profile Management Module

This module allows students and faculty members to view and edit their profile details.

5. Logout Module

This module allows users to securely exit their current session and return to the appropriate starting page.

IMPLEMENTATION:

CampusBrain is implemented as a web-based prototype using HTML, CSS, and JavaScript. HTML provides the structure of the web pages, CSS is used for styling, and JavaScript handles user interactions and application functionality.

The application provides registration and login interfaces, role-based dashboards, learning resource management, search functionality, and profile management. Students and faculty members can access features according to their selected roles. Browser storage is used to save account details and prototype data locally. The website is

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hosted using GitHub Pages, allowing users to access it through a web browser.

The implementation demonstrates the basic workflow and functionality of the CampusBrain academic resource management platform.

FUTURE SCOPE:

CampusBrain can be enhanced in several ways to improve its functionality and usefulness.

1. Secure Authentication: Implement secure user authentication to protect user accounts.

2. Centralized Database: Introduce a database to store and share learning materials across users and devices.

3. Advanced Search: Improve the search feature to help users find specific notes and study materials quickly.

4. Notifications: Add notifications for new learning materials and updates.

5. Mobile Compatibility: Improve the interface for convenient use on smartphones and tablets.

6. Additional Features: Introduce more academic tools to support learning and collaboration between students and faculty.

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CONCLUSION:

CampusBrain is a web-based prototype designed to simplify the organization and management of academic learning materials. It provides separate dashboards for students and faculty, along with features for accessing, searching, and managing study resources.

The project demonstrates the basic workflow of an academic resource management platform through a simple and user-friendly interface. Although the current version has limitations, it provides a foundation for future improvements and additional features.

Overall, CampusBrain aims to make academic resources easier to access and manage, supporting a more organized learning experience for students and faculty.

REFERENCES:

1. Mozilla Developer Network (MDN) – HTML, CSS, and JavaScript documentation.

2. GitHub Documentation – Website hosting using GitHub Pages.

3. Web Storage API Documentation – Browser storage concepts.

4. Online learning resources related to web development.

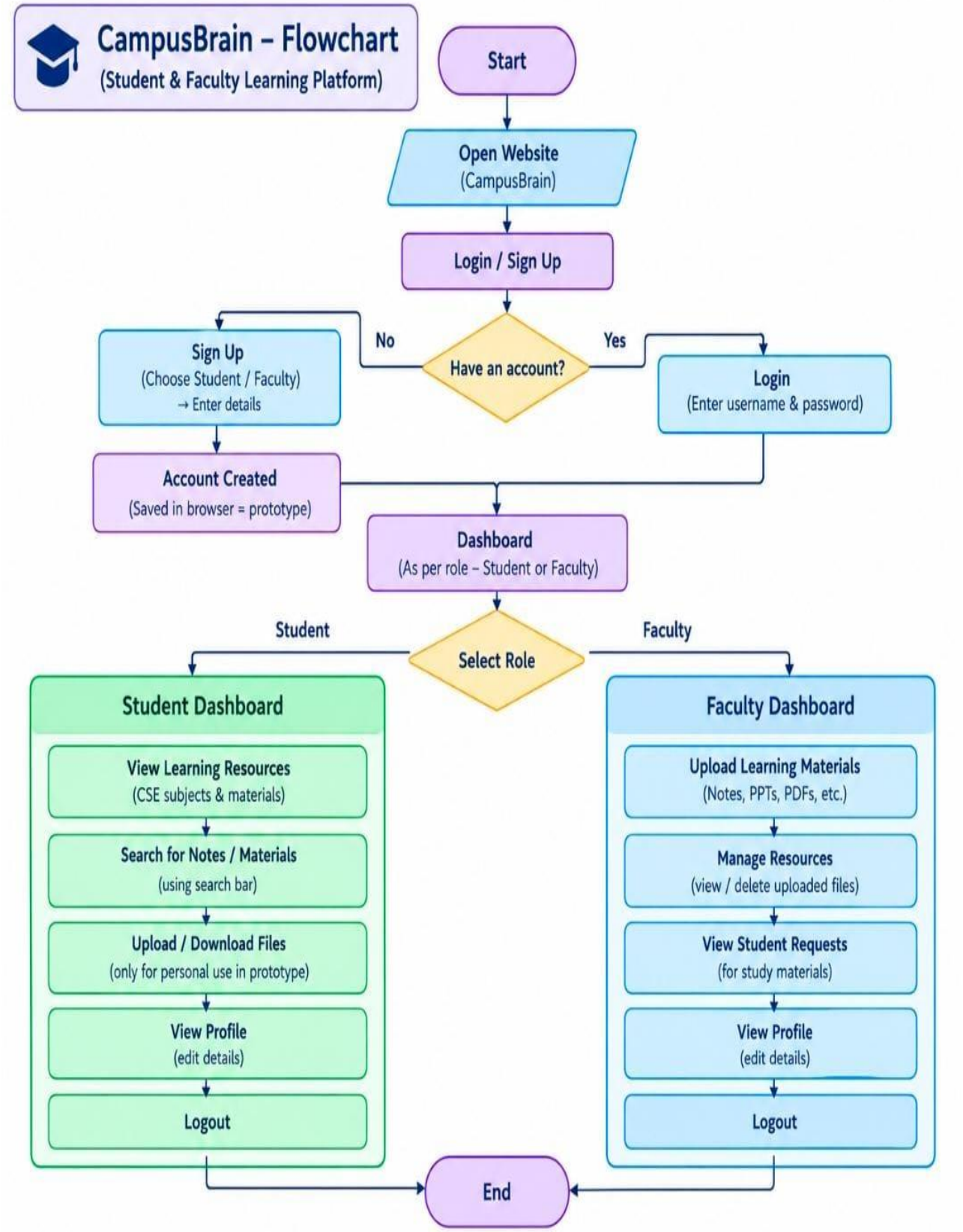
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Algorithm:

1. Start.
2. Open the CampusBrain website.
3. Select Login / Sign Up
4. Check whether the user has an account.
5. If No, select Student or Faculty, enter the details, and create an account.
6. If Yes, enter the username and password to log in.
7. Open the dashboard
8. Select the user role: Student or Faculty.
9. If the user is a Student:
   - View learning resources.
   - Search for notes and study materials.
   - Upload or download files ·View or edit the profile
10. If the user is Faculty:
   - Upload learning materials
   - Manage uploaded resources.
   - View student requests.
   - View or edit the profile.
   - Log out.
   - End.

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Flowchart:


CampusBrain – Flowchart
(Student & Faculty Learning Platform)
Start
Open Website
(CampusBrain)
Login / Sign Up
Have an account?
No
Yes
Sign Up
(Choose Student / Faculty)
→ Enter details
Login
(Enter username & password)
Account Created
(Saved in browser = prototype)
Dashboard
(As per role – Student or Faculty)
Select Role
Student
Faculty
Student Dashboard
View Learning Resources
(CSE subjects & materials)
Search for Notes / Materials
(using search bar)
Upload / Download Files
(only for personal use in prototype)
View Profile
(edit details)
Logout
Faculty Dashboard
Upload Learning Materials
(Notes, PPTs, PDFs, etc.)
Manage Resources
(view / delete uploaded files)
View Student Requests
(for study materials)
View Profile
(edit details)
Logout
End